Vid2WAM: Distilling Video Diffusion Priors into World Action Models

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Abstract

World Action Models (WAMs) improve robot policy learning by jointly modeling future visual dynamics and actions. However, their scalability and generalization remain constrained by their reliance on costly expert demonstrations. We challenge this by asking whether future supervision for WAMs must originate from target-task expert trajectories. In this paper, we propose **Vid2WAM**, an offline distillation framework that transfers visual diffusion priors from a large video foundation model into a compact WAM student. Given an observation and language instruction, Vid2WAM distills supervision through two complementary channels: task-conditioned future rollouts directly supervise the student's future prediction branch, while an inverse dynamics model recovers embodiment-specific pseudo-actions for action learning. To robustly integrate synthetic and real supervision, we introduce *source-aware residual action adaptation* that learns source-specific corrections around a shared action backbone and mitigates interference from noisy pseudo-actions. During inference, both the video teacher and inverse dynamics model are discarded, leaving only the WAM student for efficient deployment. Simulation and real-world experiments demonstrate that Vid2WAM improves novel-task generalization and data efficiency under limited expert demonstrations while preserving low-latency inference.

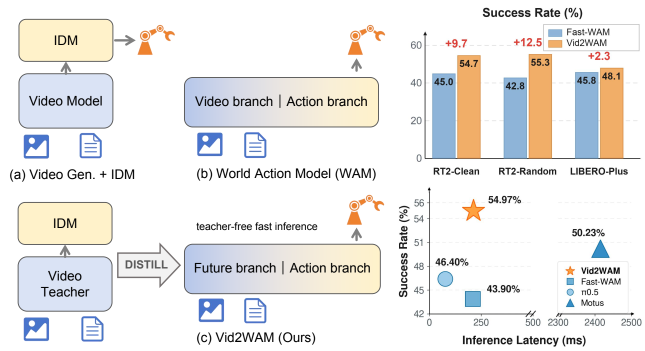

Figure 1: **Left:** Comparison of video-based robot policy paradigms. (a) Video generation plus IDM performs costly online rollout and action recovery. (b) WAM jointly learns future prediction and action generation from real video–action trajectories. (c) Vid2WAM distills teacher-generated futures and IDM pseudo-actions into a compact WAM for teacher-free inference. **Top right:** Vid2WAM consistently improves novel-task success rates over Fast-WAM. **Bottom right:** Vid2WAM achieves the highest novel-task success rate while retaining low inference latency.

Introduction

Building robot policies that generalize across tasks, objects, and environments remains a central goal of embodied intelligence. Recent progress has followed two complementary directions. Vision–language–action (VLA) models leverage Internet-scale vision–language pretraining to transfer broad semantic and linguistic knowledge into control (Zitkovich et al. 2023; Kim et al. 2025; Black et al. 2024). Meanwhile, video-based policies and World Action Models (WAMs) incorporate future prediction into policy learning, providing spatiotemporal supervision over motion, interaction, and task progression beyond static semantic supervision and instantaneous action labels (Du et al. 2023; Wu et al. 2024; Hu et al. 2025; Xu et al. 2025; Kim et al. 2026a; Ye et al. 2026).

A key advantage of WAMs stems from their future-predictive supervision. Rather than learning only from action labels, they are optimized to anticipate future visual dynamics, providing dense temporal learning signals that improve action representations even when explicit future rollouts are not required during deployment (Li et al. 2025; Xu et al. 2025; Yuan et al. 2026). However, most existing WAMs implicitly share one assumption: *future supervision must be observed in task-specific expert demonstrations*. Both future visual targets and action labels are obtained from paired robot trajectories collected through simulated data collection or teleoperation. Thus, the predictive capability of WAMs remains closely tied to scale, diversity, and cost of robot demonstrations, making it difficult to generalize beyond distributions covered by recorded target-task trajectories.

We thus argue that the assumption above is unnecessarily restrictive. Recent video foundation models have acquired remarkably rich knowledge of object motion, human-object interactions, and long-horizon visual dynamics. After adaptation with limited embodiment-specific data, they can be queried with diverse initial observations and language instructions to synthesize task-conditioned rollouts for new tasks and environments (Jang et al. 2025; Wang et al. 2026). Unlike expert demonstration collection, this process does not require a complete target-task trajectory, allowing the con-

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ditioning distribution to be expanded using simulator snapshots, inexpensive real-world observations, or controllable image synthesis. An inverse dynamics model (IDM) can further recover embodiment-specific pseudo-actions from the generated rollouts (Jang et al. 2025; Ye et al. 2025). However, treating these rollouts only as pseudo-labeled action data creates an information bottleneck: the teacher's predictive knowledge is compressed through an imperfect action-recovery model, while the generated futures themselves are discarded as direct supervision. Moreover, ambiguous transitions, generation artifacts, and physically inconsistent motion can induce systematic errors in the recovered actions (Wang et al. 2026). These limitations motivate transferring the generated rollouts through complementary visual and action channels.

This paper presents **Vid2WAM**, a framework for distilling a large video foundation model into an efficient WAM student. Given an initial observation and language instruction, an embodiment-adapted teacher generates a future rollout. We decode this rollout into pixels and re-encode it into the student's latent space, providing representation-compatible supervision for the future prediction branch. In parallel, an IDM recovers embodiment-specific pseudo-actions that supervise action learning. To mitigate the interference from pseudo-action noise, we further introduce *source-aware residual action adapters*, which provide source-specific residual corrections around a shared action backbone. At inference time, the video teacher and IDM are removed, leaving only the compact WAM student for direct action generation.

Our contributions are threefold. First, we rethink the source of future supervision for World Action Models and formulate a video-foundation-model-to-WAM distillation framework, using generated futures from video foundation models as direct supervision beyond recorded target-task expert trajectories. Second, we introduce a dual-channel, source-aware transfer strategy that jointly distills future visual dynamics and IDM-inferred actions, with residual adapters reducing interference from noisy pseudo-actions. Third, we demonstrate in simulation and real-world experiments that Vid2WAM improves expert-demonstration efficiency and novel-task generalization, while retaining efficient student-only inference.

Related Work

World Models for Robot Control. Video models provide broad priors over motion, interaction, and temporal task structure. One line of work uses these priors explicitly at deployment, generating visual subgoals or future rollouts and converting them into actions through inverse dynamics, goal-conditioned control, motion tracking, or test-time action selection (Du et al. 2023, 2024; Zhou et al. 2024; Liang et al. 2025; Collins et al. 2025; Zhou et al. 2026). Although effective for generalization, this paradigm incurs generation latency and exposes control to rollout errors. A related line of work jointly models future observations and robot actions, using future prediction to provide temporally structured supervision for policy learning (Wu et al. 2024; Cheang et al. 2024; Hu et al. 2025; Li et al. 2025; Zhu et al. 2025). Fast-WAM shows that video co-training can remain beneficial even when future generation is removed at inference (Yuan et al. 2026), while Efficient-WAM reduces cost of retaining future imagination (Li et al. 2026). These methods nevertheless derive both future and action supervision primarily from recorded robot trajectories. Vid2WAM instead decouples future supervision from target-task expert demonstrations by leveraging a video foundation model as an offline generator of task-conditioned futures, which are distilled into a compact WAM without requiring video generation at deployment.

Video-Generated Supervision for Robot Learning. Early generative augmentation methods edit objects, backgrounds, and other visual factors in recorded demonstrations while reusing their actions (Chen et al. 2023; Yu et al. 2023). More recent work also synthesizes or edits initial observations, expanding the states from which robot experience can be generated (Kim et al. 2026b). DreamGen provides the closest data-centric pipeline: it adapts a video model to a target embodiment, generates task-conditioned rollouts, and recovers pseudo-actions using an inverse dynamics or latent action model (Jang et al. 2025; Ye et al. 2025). In these pipelines, generated frames and recovered actions form synthetic demonstrations, but the generated futures do not directly supervise a student's future-prediction objective. This action-only transfer creates an information bottleneck, compressing rich predictive knowledge into imperfect pseudo-labels before it reaches the student policy. Prior work mitigates this issue by improving the executability of generated videos (Wang et al. 2026) or by avoiding synthetic action supervision and routing geometric targets to the visual backbone (Chen et al. 2026). Vid2WAM instead treats each generated rollout as dual supervision: future dynamics directly supervise world modeling, while IDM-derived actions provide embodiment-specific action grounding through source-aware residual adaptation.

Distilling Foundation-Model Priors into Robot Policies. Knowledge distillation transfers capabilities from a large teacher into a smaller student (Hinton, Vinyals, and Dean 2015), and subsequent work has extended this principle from output imitation to teacher-generated data and intermediate supervision (Hsieh et al. 2023; Mukherjee et al. 2023; Li et al. 2023). In robotics, prior methods distill planner-generated experience or foundation-model representations into deployable policies (Ha, Florence, and Song 2023; Shang et al. 2024). More closely related, S-VAM self-distills multi-step generated videos into one-step geometric and semantic foresight, while PFD distills action corrections induced by privileged recorded futures (Yan et al. 2026; Fang, Chen, and Cai 2026). CKT-WAM transfers intermediate context between heterogeneous WAMs, whereas Efficient-WAM compresses the computation required for future imagination (Jiang et al. 2026; Li et al. 2026). Vid2WAM addresses a different transfer setting. Rather than distilling predictions between robot policies or WAMs, it treats a video foundation model as an external source of future supervision. Each generated rollout provides complementary supervision for both future prediction and action learning, after which the video teacher and inverse dynamics model are discarded, leaving only the compact WAM student for deployment.

Method

Problem Formulation and Overview Given a current visual observation o_0, proprioceptive state e_0, and language instruction ℓ, we aim to learn a policy that predicts an executable action chunk $a_{1:H}$. Following the World Action Model (WAM) paradigm, the student is jointly trained on an action predictor and an auxiliary future-video predictor through a shared video-action representation.

As illustrated in Fig. 2, Vid2WAM transfers the predictive prior of a large video teacher into the student using both real demonstrations and offline teacher-generated rollouts. Real demonstrations provide expert actions and observed future-video supervision, whereas teacher rollouts provide generated future-video latents and IDM-derived pseudo-actions. The video teacher and IDM are used only offline; at deployment, the teacher, IDM, auxiliary future-video prediction head, and pseudo-source adapters are discarded, leaving the student WAM to efficiently generate action chunks from the current policy condition.

Teacher-Derived Distillation Targets

We convert the video teacher's generative prior into two complementary forms of student-compatible supervision: future-video latent targets and IDM-derived pseudo-action targets.

Video teacher adaptation. We instantiate the video teacher T_ψ with a large pretrained video diffusion model and fine-tune it using only a limited set of demonstration videos. This lightweight adaptation specializes the teacher to the robot's appearance and embodiment-specific physical constraints, while preserving the motion-prediction and instruction-following capabilities inherited from large-scale pretraining. Consequently, a small amount of in-domain demonstration data is sufficient to elicit robot-aware rollout generation. Given an initial observation o_0 and language instruction ℓ, the adapted teacher generates

$$\hat{v}_{1:T} = T_\psi(o_0, \ell). \tag{1}$$

After adaptation, the teacher is frozen and used exclusively for offline rollout generation.

Inverse dynamics model training. We train an inverse dynamics model (IDM) to recover robot motion from visual trajectories:

$$(\hat{a}_{1:H}, \hat{e}_{1:H}) = \mathrm{IDM}(\hat{v}_{1:T}), \tag{2}$$

where $\hat{a}_{1:H}$ and $\hat{e}_{1:H}$ denote the predicted action chunk and robot state sequence respectively. The IDM is supervised using action-labeled robot trajectories, but does not require language instructions or task-specific semantic annotations. It can therefore be trained using task-agnostic interaction data collected with the same robot embodiment and compatible observation setup, without requiring expert demonstrations for the target tasks. Once trained, the IDM is frozen and used only as a labeler for teacher-generated rollouts.

Pseudo-data generation and labeling. Vid2WAM keeps the real demonstration set fixed and collects an additional pool of initial states. The frozen video teacher produces visual rollouts from these initial states, which are re-encoded by the student VAE into student-space future latents $\hat{z}_{1:T}$. The IDM then recovers the corresponding pseudo-action chunks $\hat{a}_{1:H}$ and robot state sequences $\hat{e}_{1:H}$ from the generated rollouts. The resulting tuples $(o_0, \ell, \hat{z}_{1:T}, \hat{a}_{1:H}, \hat{e}_{1:H})$ are stored in an offline distillation buffer. The future latents supervise the student video branch, while the IDM-derived labels ground the generated motion in the robot action space.

Source-Aware Student Training

Vid2WAM jointly trains the student on real demonstrations and offline teacher-generated pseudo rollouts. Both domains share the video and action backbones.

Source-specific residual adaptation. Directly mixing expert actions with IDM-recovered pseudo-actions can introduce interference because the latter contain errors from video generation and inverse dynamics. We therefore place lightweight residual adapters before and after the shared action backbone, with separate parameters for the real and pseudo domains. For an action token x from domain $d \in \{\mathrm{real}, \mathrm{pseudo}\}$, an adapter computes

$$\mathrm{Ada}_d(x) = x + \alpha W^d_{\mathrm{up}} \sigma\big(W^d_{\mathrm{down}}\, \mathrm{LN}(x)\big), \tag{3}$$

where W^d_{down} and W^d_{up} form a bottleneck projection, σ is a nonlinear activation, and α controls the residual contribution. We initialize W^d_{up} to zero, so each adapter initially implements an identity mapping. The shared representation therefore remains the default path, while the residual branches provide source-dependent corrections. Only the real-domain adapter is retained at inference time.

Training objective. We use the flow-matching objective for action chunks and future-video latents (Lipman et al. 2022). Let y denote either target, we sample Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ and $t \in (0, 1)$, and construct

$$y_t = (1 - t)y + t\epsilon. \tag{4}$$

For each domain d, the model predicts the corresponding velocity field with flow-matching loss

$$\mathcal{L}_{\mathrm{FM}}(y, d) = \mathbb{E}_{y,\epsilon,t}\Big[\|f_\theta(y_t, t, c; d) - (\epsilon - y)\|_2^2\Big]. \tag{5}$$

We group the four training terms into real- and pseudo-domain objectives:

$$\begin{aligned} \mathcal{L}_{\mathrm{real}} &= \mathcal{L}_{\mathrm{FM}}(a, \mathrm{real}) + \beta \mathcal{L}_{\mathrm{FM}}(z, \mathrm{real}), \\ \mathcal{L}_{\mathrm{pseudo}} &= \mathcal{L}_{\mathrm{FM}}(\hat{a}, \mathrm{pseudo}) + \beta \mathcal{L}_{\mathrm{FM}}(\hat{z}, \mathrm{pseudo}), \end{aligned} \tag{6}$$

where (a, z) are the action and future-video targets from real demonstrations, while $(\hat{a}, \hat{z})$ are the corresponding targets from teacher-generated distillation buffers. The complete training objective is

$$\mathcal{L} = \lambda_{\mathrm{real}} \mathcal{L}_{\mathrm{real}} + \lambda_{\mathrm{pseudo}} \mathcal{L}_{\mathrm{pseudo}}, \tag{7}$$

where β balances action and video supervision, and λ_{real} and $\lambda_{\mathrm{pseudo}}$ balance the two training domains.

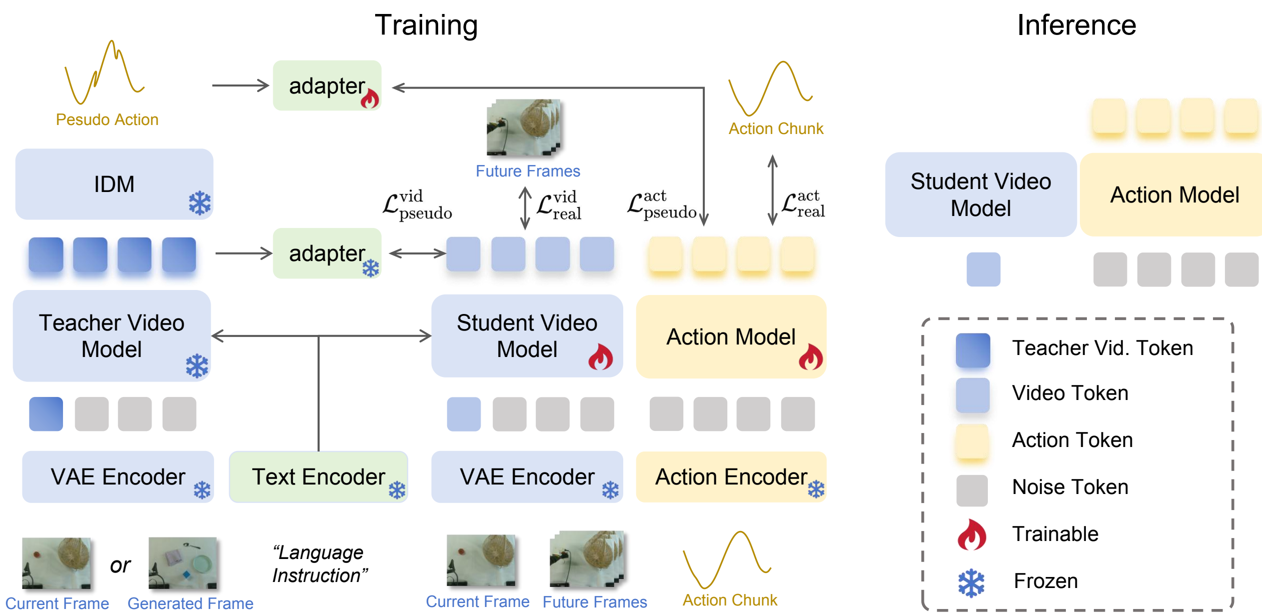

Figure 2: **Overview of Vid2WAM.** Vid2WAM distills future prediction priors from a pretrained video diffusion model into a compact World Action Model (WAM). During training, the teacher generates future videos and IDM-derived pseudo actions supervise the student WAM, while only the student is used for deployment. At inference, a single current-frame pass through the shared video backbone produces latent world features for direct action prediction.

Experiments

We investigate four questions: (1) how well existing pretrained VLA and WAM policies transfer to novel tasks; (2) whether Vid2WAM improves performance on such tasks without target-task expert trajectories; (3) whether it improves task success and perturbation robustness under a fixed, limited budget of real demonstrations; and (4) how its supervision signals and source-aware components contribute to these gains. We conduct experiments on RoboTwin 2.0 (Chen et al. 2025b), LIBERO (Liu et al. 2023), LIBERO-Plus (Fei et al. 2025), and a real-world bimanual platform.

Experimental Setup

Simulation Benchmarks. For RoboTwin 2.0, we evaluate bimanual manipulation for 50 tasks under both clean and randomized conditions, with 100 trials per task for every policy. For LIBERO, we use the Spatial, Object, Goal and Long suites, each with 10 tasks and evaluate for 50 trials per task. LIBERO-Plus stresses robustness by applying perturbations to LIBERO tasks along seven factors: camera, robot initial state, language, lighting, background, sensor noise, and layout. We use the protocol with trials on 500 variants per suite, uniformly covering all perturbations.

Real-World Setup. Our real-world platform consists of two AgileX Piper arms, one overhead Intel RealSense D435 camera, and two wrist-mounted D435 cameras. The policy takes all three views, arm joint angles and gripper states as input and predicts joint-action chunks for both arms. We evaluate six seen tasks and three held-out novel tasks spanning household and scientific manipulation. Each seen task provides 60 real trajectories collected with variations in object placement, lighting, and distractors. Each held-out task provides no real trajectory and only two true initial observations, one on clean environment with only task-specific objects and one on randomized setup with distractors. We then use GPT-Image-2 to generate 60 diverse initial-state images from the real observations for conditioning the video teacher. Because the generated images lack multi-view alignment, pseudo supervision for held-out tasks is generated from the overhead view only; the complete policy retains multi-view input when real multi-view observations are available. Detailed descriptions of real-world setup are provided in Appendix C.

Evaluation Regimes. Standard manipulation benchmarks typically train policies with expert trajectories from every evaluation task and test them under limited variations in object configurations. Such protocols primarily measure in-distribution execution and may conflate task generalization with interpolation around task-specific demonstrations. To more directly evaluate data efficiency and novel-task generalization, we consider two complementary regimes. In the **low-data regime**, each task provides only 10 expert trajectories, with Vid2WAM receiving an equal number of offline teacher-generated trajectories. In the **novel-task regime**, we withhold all expert trajectories from selected target tasks: baselines are directly transferred from the remaining tasks, whereas Vid2WAM uses only target-task instructions and initial observations to generate pseudo supervision. Neither the video teacher nor the IDM is trained on target-task trajectories. These regimes respectively evaluate policy learning under scarce real demonstrations and adaptation to novel tasks without target-task expert supervision.

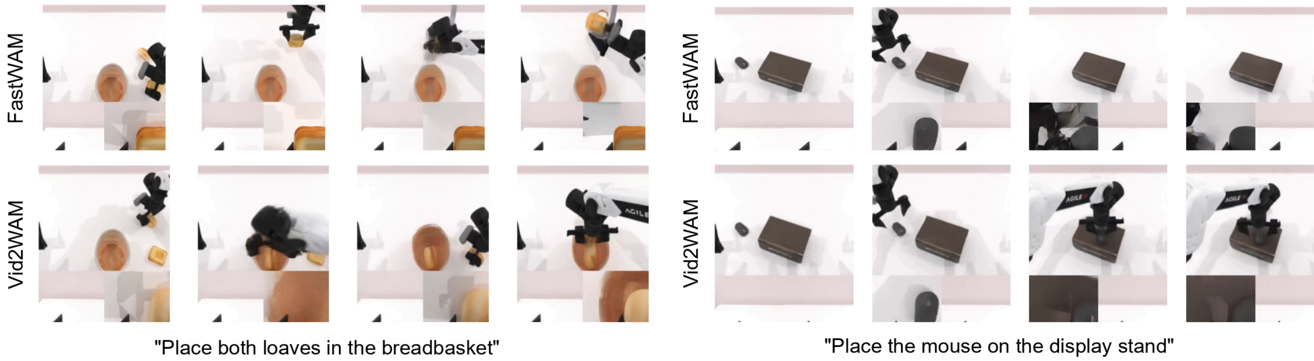

Figure 3: Qualitative future-video predictions for novel tasks. For each task, Fast-WAM (top) and Vid2WAM (bottom) receive the same initial observation and language instruction; frames progress from left to right. Fast-WAM often fails to predict coherent task progression, whereas Vid2WAM predicts future motions that more consistently reach the instructed outcome.

Figure 4: Success rates on representative novel RoboTwin tasks. Vid2WAM consistently outperforms Fast-WAM under clean and randomized evaluation.

Baselines. We compare with $\pi_{0.5}$ (Physical Intelligence et al. 2025), Motus (Bi et al. 2025), and Fast-WAM (Yuan et al. 2026). $\pi_{0.5}$ and Motus assess whether broad robot-data pretraining alone enables direct transfer to the evaluated tasks. Fast-WAM serves as the closest architectural baseline, isolating the effect of our teacher-derived future supervision and source-aware adaptation. Under each evaluation regime, all methods receive the same downstream real action-labeled demonstrations.

Implementation Details. We initialize our video teacher from the LVP checkpoint (Chen et al. 2025a), a Wan2.1-14B model (Wan et al. 2025) pretrained on robot manipulation data, and further fine-tune it on our training set. Generated frames are encoded by the student VAE so that the teacher-generated targets and student predictions lie in the same latent space. The IDM uses a ResNet-50 backbone (He et al. 2016) with action and proprioception prediction heads. To reduce the influence of long-horizon generation errors, pseudo supervision in simulation experiments is restricted to the first eight seconds of each generated rollout. For real-world novel-task pseudo data, the spatial video loss is computed only over the valid overhead-view region, while pseudo-action supervision remains unchanged. Complete training schedules, IDM validation results, multi-view input layouts, and spatial and temporal masking details are provided in Appendix A& C.

Method	Low-data		Novel regime		Novel subset	
	Clean	Rand.	Clean	Rand.	Clean	Rand.
$\pi_{0.5}$	48.7	42.8	67.9	66.5	46.5	46.3
Motus	49.9	43.6	75.1	76.3	48.9	51.5
Fast-WAM	52.1	39.1	75.7	74.7	45.0	42.8
Ours	**55.4**	**45.4**	**78.3**	**78.5**	**54.7**	**55.3**

Table 1: Overall success rates (%) on RoboTwin 2.0. Novel regime averages over 35 seen and 15 unseen tasks, while Novel subset reports the average over 15 unseen tasks only.

Simulation Results

RoboTwin 2.0. Table 1 reports results under clean and randomized conditions on RoboTwin 2.0, including averages over all 50 tasks and over the 15 novel tasks in the novel regime. The results reveal that both pretrained VLA policies and WAMs struggle to transfer to novel tasks, with all baselines dropping substantially on the 15-task novel subset. This degradation is particularly pronounced for Fast-WAM, whose success rate decreases from 75.7% to 45.0% under clean evaluation and from 74.7% to 42.8% under randomiza-

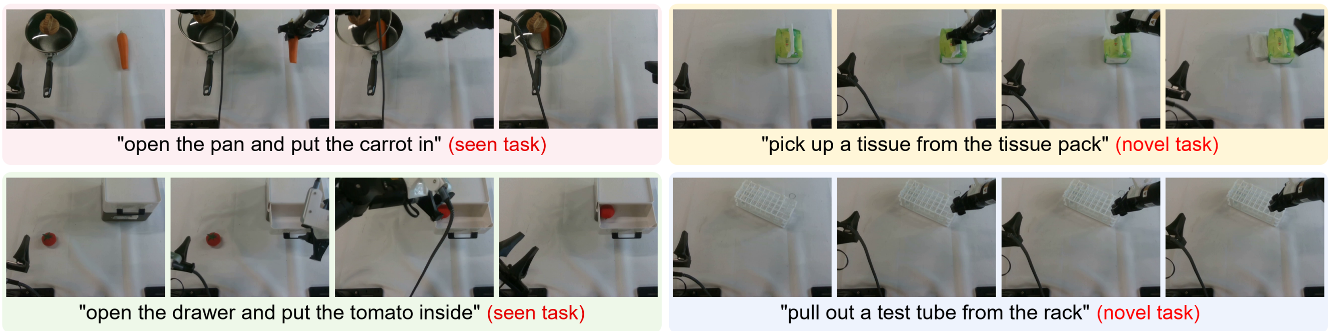

Figure 5: Vid2WAM real-world executions for representative seen and novel tasks. Frames progress left to right.

Regime	Model	Suite				Perturbation							Overall
		Spatial	Object	Goal	Long	Cam.	Robot	Lang.	Light	BG	Noise	Layout	
Low-data	Motus	34.2	35.4	**29.4**	12.4	2.5	29.5	53.3	38.1	31.0	6.0	38.2	27.9
	Fast-WAM	39.8	54.0	14.6	**25.8**	**8.5**	22.5	49.5	57.8	32.1	23.3	45.6	33.6
	Ours	**50.0**	**55.0**	27.8	23.0	8.2	**31.2**	**57.9**	**71.6**	**36.8**	**23.7**	**48.8**	**39.0**
Novel	Motus	52.2	44.0	31.8	**39.6**	4.7	**51.6**	66.3	63.8	**46.2**	18.7	47.7	41.9
	Fast-WAM	48.0	59.6	37.0	38.6	24.1	42.5	64.2	67.2	39.0	34.6	**52.7**	45.8
	Ours	**52.4**	**60.2**	**41.4**	38.6	**28.5**	42.8	**67.7**	**67.5**	39.7	**42.4**	51.6	**48.1**

Table 2: Average task success rates (%) on LIBERO-Plus under the low-data and novel regimes, evaluated across four suites and seven perturbation types.

Method/Regime	Spatial	Object	Goal	Long	**Avg.**
Low-data					
$\pi_{0.5}$	91.2	95.6	84.4	76.4	86.9
Motus	**93.0**	94.4	85.0	78.4	87.7
Fast-WAM	89.6	97.8	82.8	78.8	87.3
Ours	**93.0**	**98.6**	**87.6**	**79.4**	**89.7**
Novel					
$\pi_{0.5}$	77.2	**80.2**	78.4	**75.8**	77.9
Motus	77.4	72.2	68.8	69.8	72.1
Fast-WAM	76.8	79.0	77.8	73.0	76.7
Ours	**79.4**	79.4	**79.2**	75.2	**78.3**

Table 3: Average task success rates (%) across the four LIBERO suites under the low-data and novel regimes.

tion. By distilling the pretrained video teacher's predictive prior, Vid2WAM improves novel-subset success to 54.7% and 55.3%, outperforming Fast-WAM by 9.7 and 12.5 points. Vid2WAM also achieves the best performance in the low-data regime, demonstrating consistent benefits for both novel-task transfer and learning from limited real demonstrations.

Figure 4 presents representative novel tasks for which Vid2WAM clearly improves over Fast-WAM. The teacher-generated future trajectories provide useful information about task completion and guide the policy toward the intended outcome. Figure 3 further compares the predicted future videos produced from identical initial observations and language instructions. Fast-WAM often stops after an intermediate interaction or loses visual coherence, while Vid2WAM continues toward the requested placement. The difference becomes most visible after the policy has completed an opening or grasping step, suggesting that the distilled future information supports longer-horizon task progression.

LIBERO and LIBERO-Plus. Table 3 reports results on LIBERO under both the low-data and novel regimes. In the low-data regime, Vid2WAM achieves an average success rate of 89.7%, obtaining the best or tied-best result on every suite. In the novel regime, it also achieves the highest average success rate among the four evaluated methods. These results show that Vid2WAM improves performance both when expert demonstrations are limited and when no real trajectories are available for the target tasks.

Table 2 further evaluates robustness under the perturbations introduced by LIBERO-Plus. Vid2WAM achieves the highest overall performance on both regimes, with particularly clear gains under robot-state, language, and lighting perturbations on the low-data regime. We further evaluate whether the novel regime results depend on the selected target tasks. We consider three LIBERO or LIBERO-Plus splits that withhold novel task IDs $0/1$, $3/6$, or $8/9$ from each suite, where the $8/9$ split corresponds to the settings in Table 2 and Table 3. Results in Appendix B show that Vid2WAM always achieves a higher average success rate than Fast-WAM on all three splits in both benchmarks. Results in Appendix B also evaluate only the novel tasks from these LIBERO and LIBERO-Plus splits, and Vid2WAM again outperforms Fast-WAM on every split, showing that the aggregate im-

Regime	Task	$\pi_{0.5}$	Motus	Fast-WAM	Ours
Seen	Click Bell	30%	**70%**	20%	65%
	Item Handover	25%	**60%**	25%	**60%**
	Carry Basket	45%	45%	20%	**50%**
	Open Pan Place	5%	**40%**	10%	**40%**
	Open Drawer Place	60%	55%	30%	**65%**
	Insert Test Tube	15%	0%	10%	**25%**
Novel	Pick Test Tube	0%	10%	5%	**30%**
	Take Tissue	0%	0%	5%	**15%**
	Tomato Basket	0%	20%	25%	**30%**

Table 4: Real-world bimanual task success rates.

provements result from better adaptation to novel tasks rather than the selection of some specific tasks.

Real-World Experiments

We evaluate real-world performance on nine contact-rich manipulation tasks, covering both seen and novel settings. Seen tasks include *Click Bell*, *Item Handover*, *Carry Basket*, *Open Pan Place*, *Open Drawer Place* and *Insert Test Tube*. They use real trajectories during training. Novel tasks include *Pick Test Tube*, *Take Tissue* and *Tomato Basket*. They are designed to test behavior-level or object-level novelty, requiring novel manipulation actions or new objects that are absent from the real expert trajectories. These tasks use initial observations (true and synthetic) and teacher-generated supervision rather than real demonstrations. Detailed real-world task descriptions are provided in Appendix C.

Table 4 reports success rates on 20 real-robot trials for all models on each task. Among all tasks, Vid2WAM achieves the best or competitive performance across the four models. Figure 5 further visualizes representative real-world executions of Vid2WAM on seen and novel tasks. In the novel task, the robot can extract a deformable tissue or pull out a transparent test tube despite having no real action-labeled trajectories.

Ablation Studies

Table 5 evaluates on different variants of our Vid2WAM under LIBERO low-data regime. *Dual Action DiT* variant uses separate action DiTs for ground-truth and teacher-generated samples while sharing the video DiT; *No Adapter* mixes both sources in shared action and video DiTs. *Pseudo Action Only* retains Vid2WAM's synthetic rollouts and adapters but removes the latent-video objective, while *Future Latent Only* removes pseudo-action supervision and uses only future latents for distillation.

Results show that among the dual-channel variants, independent action DiTs outperform fully shared training, while the full adapter design (Ours) performs best on average. *Future Latent Only* outperforms *Pseudo Action Only* by 1.8 points on average and Fast-WAM by 0.9 points, showing that future latents provide a useful guidance signal. Its lower average than our full Vid2WAM further suggests that pseudo actions provide complementary supervision.

We additionally evaluate an online *Teacher Policy* that composes the video teacher and IDM at inference time without student distillation. It underperforms Fast-WAM and all distilled variants, showing that the gains come from offline future supervision combined with ground-truth action anchoring rather than a stronger online teacher controller. This result is also consistent with generation and IDM errors accumulating under task execution. Together with the *No Adapter* ablation, it suggests that source-dependent residual corrections can mitigate the influence of noisy pseudo-actions.

Method	Spatial	Object	Goal	Long	**Avg.**
Fast-WAM	89.6	97.8	82.8	78.8	87.3
Ours	93.0	**98.6**	**87.6**	79.4	**89.7**
Dual Action DiT	92.2	97.8	85.8	**80.6**	89.1
No Adapter	**93.8**	97.6	84.2	78.4	88.5
Pseudo Action Only	90.4	96.0	85.0	74.0	86.4
Future Latent Only	92.8	98.2	86.4	75.2	88.2
Teacher Policy	76.0	85.2	65.6	58.2	71.3

Table 5: Ablation study on LIBERO benchmark (low-data regime).

Method	Params	Mean	Std.	Min	Max
Fast-WAM	5B	209.1	4.7	204.8	220.2
Ours	5B	212.5	11.7	199.0	227.8
Teacher Policy	14B	4894.5	106.6	4785.8	4998.9
$\pi_{0.5}$	3.3B	76.7	0.4	76.2	77.6
Motus	8B	2414.9	22.8	2380.0	2456.3

Table 6: Inference latency (ms) on a single RTX 4090 GPU.

Inference Latency Analysis

Table 6 reports chunk-level inference latency on a single RTX 4090 GPU, excluding cold-start costs. Results show that our Vid2WAM approximately matches Fast-WAM in latency, and is substantially faster than Motus and the Teacher Policy, though slower than $\pi_{0.5}$.

Conclusion

We present Vid2WAM, a framework for transferring the predictive diffusion priors of a large video model into a compact world-action policy. By using teacher-generated futures and IDM-inferred actions as offline supervision, Vid2WAM enables the student to acquire behavioral knowledge beyond the available expert trajectories, while source-aware adaptation limits the impact of noisy synthetic action labels. Across simulation and real-world manipulation, Vid2WAM improves data efficiency and generalization to novel tasks without introducing teacher inference at deployment. Our findings support a broader paradigm in which large pretrained models serve as offline knowledge sources for efficient robot policies. Video generators, VLMs, and other multimodal foundation models could supervise compact policies through predicted subgoals, future representations, or action-conditioned outcomes, offering a scalable path for transferring their predictive capabilities to efficient robot policies.

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A Implementation Details

For all models including video teacher, IDM and Vid2WAM, on LIBERO and LIBERO-Plus, the head and wrist camera views are resized to 224×224 and concatenated horizontally into a 224×448 frame. For RoboTwin 2.0, the top head camera is resized to 256×320, while the two wrist views are resized to 128×160, concatenated horizontally, and placed below to form a 384×320 canvas. For real-world experiments, the video teacher generates overhead-view rollouts at its native 480×640 resolution. For student training, these frames are resized to occupy the upper 224×448 region of a 448×448 composite, while the two unavailable 224×224 wrist-view slots in the lower half are black-padded. For real multi-view samples, the true wrist views occupy these lower slots.

Teacher and Inverse Dynamics Model Training

Video teacher training. The teacher is an image-to-video diffusion model initialized from the Large Video Planner checkpoint (Chen et al. 2025a), which adopts the Wan2.1-I2V-14B architecture (Wan et al. 2025). We fine-tune the diffusion transformer while keeping the UMT5 text encoder, CLIP image encoder, and video VAE frozen. All video teachers are trained for around 500 epochs. Each training sample consists of 49 video frames, which are encoded into 16-channel latent representations using the frozen Wan VAE.

Given a clean video latent z, Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$, and $t \sim \mathcal{U}(0,1)$, we apply the time shift

$$\tau = \frac{st}{1 + (s-1)t}, \qquad s = 3, \tag{S1}$$

and construct

$$z_\tau = (1-\tau)z + \tau\epsilon. \tag{S2}$$

The teacher predicts the flow target $v^\star = \epsilon - z$ and is optimized with

$$\mathcal{L}_{\text{teacher}} = \mathbb{E}_{z,\epsilon,t}\left[\|v_\Theta(z_\tau, \tau, c) - (\epsilon - z)\|_2^2\right], \tag{S3}$$

where c contains the language embedding and first-frame image condition. We additionally use diffusion forcing (Chen et al. 2024) during training. In particular, one or two latent history frames are sampled with probabilities 0.9 and 0.1, respectively, and the history is kept clean with probability 0.7.

Inverse dynamics model. The inverse dynamics model (IDM) maps a short video window to the action at its center frame. Given three observations separated by a domain-dependent stride δ, it predicts

$$\widehat{a}_t = q_\psi\left(I_{t-\delta}, I_t, I_{t+\delta}\right). \tag{S4}$$

We use an ImageNet-pretrained ResNet-50 (He et al. 2016) whose first convolution is expanded from three to nine input channels to jointly process the three RGB frames. The pretrained convolutional filters are repeated across the temporal dimension and divided by three. The final ResNet feature map is passed through a learnable spatial-softmax layer and a lightweight MLP action head.

Hyperparameter	Value
Initialization	Wan2.1-I2V-14B-480P with LVP 14B weights
Layers	40
Hidden dimension	5120
Feed-forward dimension	13824
Attention heads	40
Training frames	49
VAE latent channels	16
VAE temporal/spatial stride	$4 \times 8 \times 8$
Continuous-time shift	3.0
Training-time scale	1000
Optimizer	AdamW
Learning rate	8×10^{-6}
AdamW betas	$(0.9, 0.95)$
Weight decay	5×10^{-2}
Warm-up steps	100
Batch size	4
Gradient accumulation	4
Numerical precision	BF16
Maximum text length	512

Table S1: Shared hyperparameters used to fine-tune the video teacher.

Hyperparameter	Value
Visual backbone	ResNet-50
Number of input frames	3
Input channels	9
Feature pooling	Spatial softmax
Hidden activation	GELU
Dropout	0.1
Training loss	Smooth L1
Optimizer	AdamW
Learning rate	3×10^{-4}
AdamW betas	$(0.9, 0.999)$
Weight decay	10^{-2}
Training iterations	200,000
Batch size	32
Learning-rate schedule	Cosine
Warm-up iterations	5,000
Final learning-rate ratio	0
Numerical precision	BF16

Table S2: Shared training hyperparameters for the action IDM.

The action mean μ_a and standard deviation σ_a are computed from the training split. The IDM is trained with a Smooth L1 loss in the normalized action space:

$$\mathcal{L}_{\text{IDM}} = \mathbb{E}\left[\text{SmoothL1}\left(\frac{\widehat{a}_t - \mu_a}{\sigma_a}, \frac{a_t - \mu_a}{\sigma_a}\right)\right]. \tag{S5}$$

The prediction is converted back to the original action space during inference.

IDM training and validation. For LIBERO/LIBERO-Plus, RoboTwin and real-world, we train separate ResNet-50 IDMs for action and proprioceptive-state targets. Each IDM is trained for 200,000 steps using compatible action-labeled trajectories; withheld-task trajectories are excluded.

Domain	Resolution	FPS	Frames	Steps	Language CFG	History CFG
LIBERO	224×448	15	49	40	3.0	1.0
RoboTwin	384×320	15	49	40	3.0	1.0
Real-world	480×640	10	49	40	3.0	2.0

Table S3: Domain-specific video-teacher rollout configurations used for constructing the pseudo-data buffers.

Dataset	IDM Target	Val Loss ↓	Val Acc ↑	Val L1 ↓
LIBERO / LIBERO-Plus	Action	0.0054	85.8%	0.0112
LIBERO / LIBERO-Plus	State	0.0000	100.0%	0.0020
RoboTwin	Action	0.0009	81.9%	0.0100
RoboTwin	State	0.0008	83.1%	0.0095
Real-world	Action	0.0002	98.1%	0.0044
Real-world	State	0.0002	98.6%	0.0043

Table S4: Best validation performance of the trained IDM models.

The frozen IDMs then label pseudo-actions from teacher-generated videos.

For validation, we randomly partition the frame-level samples with size N. The validation-set size is

$$N_{\text{val}} = \min\left(\lfloor 0.05N \rfloor, 2000\right), \tag{S6}$$

with at least one validation sample. Target normalization statistics are computed exclusively from the resulting training split. We evaluate the IDM every 5,000 optimization steps and retain the checkpoint with the lowest validation Smooth L1 loss. We further define validation accuracy as the proportion of samples for which all predicted dimensions fall within their corresponding absolute-error tolerances:

$$\text{ValAcc} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\bigcap_{j=1}^{D} \{|\hat{y}_{i,j} - y_{i,j}| < \tau_j\}\right), \tag{S7}$$

where N is the number of validation samples, D is the target dimensionality, $\hat{y}_{i,j}$ and $y_{i,j}$ denote the prediction and ground-truth value for dimension j of sample i respectively, and $\tau_j = 0.05$ for every dimension. Consequently, a sample is counted as correct only if every predicted dimension satisfies the tolerance. The rules are identical for action and state IDM training.

Table S4 reports validation results for all IDMs, with all models achieving higher than 80% accuracy, especially both real-world heads exceeding 98% accuracy.

Pseudo-data construction. After training, the video teacher and IDM are used only as offline data generators. Table S3 lists the rollout parameters used to construct each distillation buffer. Rollout generation takes only the source episode's first visual observation together with the language instruction, after which the future observations are synthesized by the video teacher and the pseudo actions are recovered by the IDM.

Because the Wan2.1 teacher VAE uses 16 latent channels while the Vid2WAM VAE uses 48, teacher rollouts are decoded to RGB and re-encoded by the frozen student VAE before storage.

The 49 frames in Table S3 denote one generation chunk, not a complete pseudo-rollout. Successive chunks are assembled into longer videos before the simulation cutoff and pseudo-data construction.

Teacher-generated rollout examples. Figure S1 shows representative teacher rollouts across LIBERO, RoboTwin, and real-world settings. The examples cover both low-data and novel-task regimes, as well as clean and randomized real-world initial states.

Across the six examples, the teacher predicts task-relevant object motion and multi-step manipulation progress while maintaining the scene context needed for the instruction. The novel-task and randomized examples further illustrate useful future-progress cues for held-out instructions and perturbed initial states.

Data Budget and Model Exposure

Table S5 reports the data budget. For all experiment setups, the video teachers and IDMs are fine-tuned on its regime-specific dataset: the low-data regime uses corresponding expert demonstrations, while the novel regime excludes held-out-task trajectories.

Vid2WAM Training Details

Model and distillation training details. Table S6 summarizes the model architecture and training hyperparameters used by Vid2WAM. Each iteration processes one real batch and one pseudo batch before a single parameter update. Thus, the reported global batch size of 128 per domain denotes 128 real and 128 pseudo samples per update.

For LIBERO/LIBERO-Plus and real-world, we trained Vid2WAM for 10 epochs, while for RoboTwin 2.0, we trained for 5 epochs.

Details on ablation variants. Details on the Vid2WAM-based ablation variants discussed in the main paper are shown in Table S7. All Vid2WAM-based variants share the same teacher and IDM models and corresponding distillation buffers. Training hyperparameters also remain the same.

For the Teacher Policy described in the main paper, we also use the same teacher and IDM models as Vid2WAM across all regimes and benchmarks. This policy uses the teacher to generate task rollouts and the IDM to infer action trajectories for direct execution.

B Sensitivity to Novel-Task Selection

The split reported in the main paper withholds task IDs $8/9$ from every LIBERO and LIBERO-Plus suite; we additionally evaluate $0/1$ and $3/6$. Tables S11 and S10 report full-benchmark performance, whereas Tables S12 and S13 evaluate only the withheld tasks.

Task-ID Mapping

Task IDs are suite-local, and LIBERO-Plus uses the same mapping as LIBERO. Tables S8 and S9 list the task-ID mappings for all four suites. A split withholds corresponding tasks with specified IDs within each suite.

Figure S1: Representative teacher-generated rollouts across LIBERO, RoboTwin, and real-world settings. Each row presents six temporally ordered keyframes conditioned on the displayed language instruction. The simulation examples include low-data and novel-task regimes, while the real-world examples cover clean and randomized initial states.

From Table S11 and Table S10, Vid2WAM improves the average LIBERO and LIBERO-Plus success rate for all three splits, which shows that Vid2WAM's enhancement over Fast-WAM on the novel regime does not depend on the selected target tasks.

Withheld-Task Results

Tables S12 and S13 isolate the withheld target tasks in LIBERO and LIBERO-Plus rather than averaging over the full benchmark suites.

Benchmark/regime	Tasks		Expert trajectories (seen tasks only)	Teacher rollouts	Held-out-task exposure	
	Seen	Novel			Teacher	IDM
LIBERO low-data	40	0	400 (10 per task)	400	/	/
LIBERO novel	32	8	1,366	312	No	No
RoboTwin low-data	50	0	1000 (10 clean + 10 randomized per task)	1000	/	/
RoboTwin novel	35	15	19250 (50 clean +500 randomized per seen-task)	8250	No	No
Real-world	6	3	360 (60 per seen-task)	180	No	No

Table S5: All data budgets and exposures. "Teacher exposure" and "IDM exposure" indicate whether the corresponding model was trained on trajectories from the evaluated held-out target tasks; this is distinct from using a trajectory's first frame to seed a rollout.

Hyperparameter	Value
Video backbone initialization	Wan2.2-TI2V-5B
Number of transformer layers	30
Number of attention heads	24
Attention head dimension	128
Video hidden dimension	3072
Video FFN dimension	14336
Action hidden dimension D	1024
Action FFN dimension	4096
Text embedding dimension	4096
Maximum text length	128
Video patch size	$(1, 2, 2)$
Bottleneck dimension D_r	128
Adapter activation	GELU
Adapter dropout	0.0
Adapter residual scale α	1.0
Up-projection initialization	Zero
Flow-matching training steps	1000
Video noise-schedule shift	5.0
Action noise-schedule shift	5.0
Balance coefficient β	1.0
λ_{real}	1.0
λ_{pseudo}	0.25
Global batch per domain	128
Gradient accumulation	1
Learning rate	LIBERO/RoboTwin: 1×10^{-4}; real-world: 5×10^{-5}
LR schedule	Cosine
Warmup ratio	5%
Minimum LR ratio	0.01
Optimizer	AdamW
AdamW β_1, β_2	$(0.9, 0.95)$
Weight decay	0.01
Maximum gradient norm	1.0
Numerical precision	BF16

Table S6: Model and action-adapter hyperparameters in Vid2WAM.

Novel IDs	Method	Spatial	Object	Goal	Long	**Avg.**
$0/1$	Fast-WAM	29.0	1.0	0.0	0.0	7.5
	Vid2WAM	**90.0**	**63.0**	0.0	0.0	**38.3**
$3/6$	Fast-WAM	0.0	0.0	0.0	0.0	0.0
	Vid2WAM	**1.0**	**26.0**	**1.0**	0.0	**7.0**
$8/9$	Fast-WAM	0.0	0.0	0.0	0.0	0.0
	Vid2WAM	**2.0**	**2.0**	**4.0**	**1.0**	**2.3**

Table S12: Success rates (%) evaluated only on the withheld LIBERO target tasks. Each split removes the listed two task IDs from every suite.

Results show that Vid2WAM improves average withheld-target success for all three splits on both benchmarks. The "zero to one" improvements on most suites indicate that our distillation framework can adjust the task execution process correctly. Low absolute success on several Goal and Long targets nevertheless shows that adaptation without target-task action trajectories remains somewhat difficult.

Variant	Structural or objective change	Training-time parameter change	Inference path
Vid2WAM (default)	Shared video expert and shared ActionDiT; separate input/output adapters for both domains.	around +1M	Shared ActionDiT with the real-domain adapters.
Dual Action DiT	Shared video expert, but independent real and pseudo Action DiTs.	around +1B	Real ActionDiT only
No Adapter	One shared video expert and one shared Action DiT, with no domain-specific adapter modules. Both domains update the same path.	0	The shared Action DiT path.
Pseudo-Action Only	Discard the future latent flow-matching loss.	Same as Vid2WAM	Same as Vid2WAM
Future-Latent Only	Discard the pseudo-action flow-matching loss.	Same as Vid2WAM	Same as Vid2WAM

Table S7: Exact implementation of the distillation ablations.

ID	LIBERO-Spatial instruction	ID	LIBERO-Object instruction
0	pick up the black bowl between the plate and the ramekin and place it on the plate	0	pick up the alphabet soup and place it in the basket
1	pick up the black bowl from table center and place it on the plate	1	pick up the bbq sauce and place it in the basket
2	pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate	2	pick up the butter and place it in the basket
3	pick up the black bowl next to the cookie box and place it on the plate	3	pick up the chocolate pudding and place it in the basket
4	pick up the black bowl next to the plate and place it on the plate	4	pick up the cream cheese and place it in the basket
5	pick up the black bowl next to the ramekin and place it on the plate	5	pick up the ketchup and place it in the basket
6	pick up the black bowl on the cookie box and place it on the plate	6	pick up the milk and place it in the basket
7	pick up the black bowl on the ramekin and place it on the plate	7	pick up the orange juice and place it in the basket
8	pick up the black bowl on the stove and place it on the plate	8	pick up the salad dressing and place it in the basket
9	pick up the black bowl on the wooden cabinet and place it on the plate	9	pick up the tomato sauce and place it in the basket

Table S8: Suite-local LIBERO-Spatial and LIBERO-Object task-ID mappings, shared by LIBERO and LIBERO-Plus.

ID	LIBERO-Goal instruction	ID	LIBERO-Long instruction
0	open the middle drawer of the cabinet	0	turn on the stove and put the moka pot on it
1	open the top drawer and put the bowl inside	1	put the black bowl in the bottom drawer of the cabinet and close it
2	push the plate to the front of the stove	2	put the yellow and white mug in the microwave and close it
3	put the bowl on the plate	3	put both moka pots on the stove
4	put the bowl on the stove	4	put both the alphabet soup and the cream cheese box in the basket
5	put the bowl on top of the cabinet	5	put both the alphabet soup and the tomato sauce in the basket
6	put the cream cheese in the bowl	6	put both the cream cheese box and the butter in the basket
7	put the wine bottle on the rack	7	put the white mug on the left plate and put the yellow and white mug on the right plate
8	put the wine bottle on top of the cabinet	8	put the white mug on the plate and put the chocolate pudding to the right of the plate
9	turn on the stove	9	pick up the book and place it in the back compartment of the caddy

Table S9: Suite-local LIBERO-Goal and LIBERO-Long task-ID mappings, shared by LIBERO and LIBERO-Plus.

C Real-World Experiment Details

Figure S2 illustrates the real-world platform used for data collection and evaluation. A camera-equipped dual-arm robot is operated through the paired teleoperation interface. The top-down camera, studio lighting, and reconfigurable tabletop provide controlled variation in viewpoint, illumination, texture, object layout, and task composition.

Novel IDs	Model	Suite				Perturbation							Overall
		Spatial	Object	Goal	Long	Cam.	Robot	Lang.	Light	BG	Noise	Layout	
0/1	Fast-WAM	58.4	61.2	**39.4**	37.4	**28.2**	36.1	60.0	70.5	**54.5**	39.6	**58.7**	49.1
	Vid2WAM	**60.2**	**63.2**	37.2	**39.8**	24.8	**40.0**	**66.3**	**76.5**	48.4	**40.6**	**58.7**	**50.1**
3/6	Fast-WAM	40.6	**56.6**	30.2	**36.4**	13.2	37.5	52.3	64.2	**43.0**	**32.9**	48.4	41.0
	Vid2WAM	**45.0**	53.6	**40.2**	35.0	**15.1**	**38.3**	**63.2**	**70.9**	40.1	31.8	**49.8**	**43.5**
8/9	Fast-WAM	48.0	59.6	37.0	**38.6**	24.1	42.5	64.2	67.2	39.0	34.6	**52.7**	45.8
	Vid2WAM	**52.4**	**60.2**	**41.4**	**38.6**	**28.5**	**42.8**	**67.7**	**67.5**	**39.7**	**42.4**	51.6	**48.1**

Table S10: Sensitivity to novel-task selection on LIBERO-Plus. Each split treats the listed two task IDs in every suite as novel; 8/9 is the main novel setting reported in the main paper. Best results within each split are bolded.

Novel IDs	Method	Spatial	Object	Goal	Long	**Avg.**
0/1	Fast-WAM	82.4	79.8	**76.4**	**75.6**	78.6
	Vid2WAM	**94.2**	**92.6**	76.2	71.8	**83.7**
3/6	Fast-WAM	73.0	79.2	**77.6**	**76.4**	76.6
	Vid2WAM	**75.6**	**84.4**	76.2	74.2	**77.6**
8/9	Fast-WAM	76.8	79.0	77.8	73.0	76.7
	Vid2WAM	**79.4**	**79.4**	**79.2**	**75.2**	**78.3**

Table S11: Sensitivity to novel-task selection on LIBERO. Best results within each split are bolded.

Novel IDs	Method	Spatial	Object	Goal	Long	**Avg.**
0/1	Fast-WAM	51.0	1.0	0.0	0.0	13.0
	Vid2WAM	**69.0**	**34.0**	0.0	0.0	**25.8**
3/6	Fast-WAM	0.0	0.0	1.0	0.0	0.3
	Vid2WAM	**2.0**	**9.0**	**2.0**	0.0	**3.3**
8/9	Fast-WAM	1.0	2.0	1.0	0.0	1.0
	Vid2WAM	**2.0**	**12.0**	**5.0**	0.0	**4.8**

Table S13: Success rates (%) evaluated only on the withheld LIBERO-Plus target tasks. Each split removes the listed two task IDs from every suite.

Detailed Task Description

The evaluation includes six seen tasks with expert trajectories and three novel tasks trained only from true or synthetic initial observations, language instructions, and offline teacher supervision. The novel tasks introduce behaviors or object–action compositions absent from the labeled trajectories.

Seen tasks.

- **Click Bell:** the right gripper closes and presses a tabletop bell.
- **Item Handover:** the right arm lifts a pen upright, transfers it to the left arm, and the left arm drops it into a pen holder.
- **Carry Basket:** the left arm places a milk carton into a basket, after which the right arm lifts and relocates the basket by its handle.
- **Open Pan Place:** the left arm opens the pan lid, the right arm places a carrot inside, and the left arm closes the lid.
- **Open Drawer Place:** the right arm opens a handled drawer while the left arm places a tomato inside.
- **Insert Test Tube:** the arms hand over a transparent test tube before vertically inserting it into a rack.

Novel tasks.

- **Pick Test Tube:** the right arm grasps a test tube already inserted in the rack and pulls it out.
- **Take Tissue:** the right arm pinches and extracts a tissue from a soft tissue pack.
- **Tomato Basket:** the left arm picks up a tomato and place it into the basket.

Initial-State Augmentation

As shown in Figure S3, for each novel task, GPT-Image-2 generates 30 synthetic images from a collected overhead observation in clean setting with only task-specific objects, and 30 images from an observation with different distractors. The generated images vary object placement and illumination without requiring additional real frames.

Spatial and Temporal Masking

Synthesis is restricted to the overhead view because independently generated views do not reliably preserve cross-view alignment. To fit the model, the generated image occupies the upper camera slot, while unavailable wrist-camera regions are zero-filled, and a spatial latent mask limits pseudo-video loss to the valid region. Ground-truth multi-view and action supervision remain unmasked.

Spatial masking. In the single-view construction, the overhead image is resized to the top 224×448 region of a 448×448 composite image, while the bottom 224×448

Figure S2: Real-world bimanual platform, including the teleoperation interface, camera-equipped dual-arm robot, top-down camera, studio lighting, and reconfigurable tabletop with task objects.

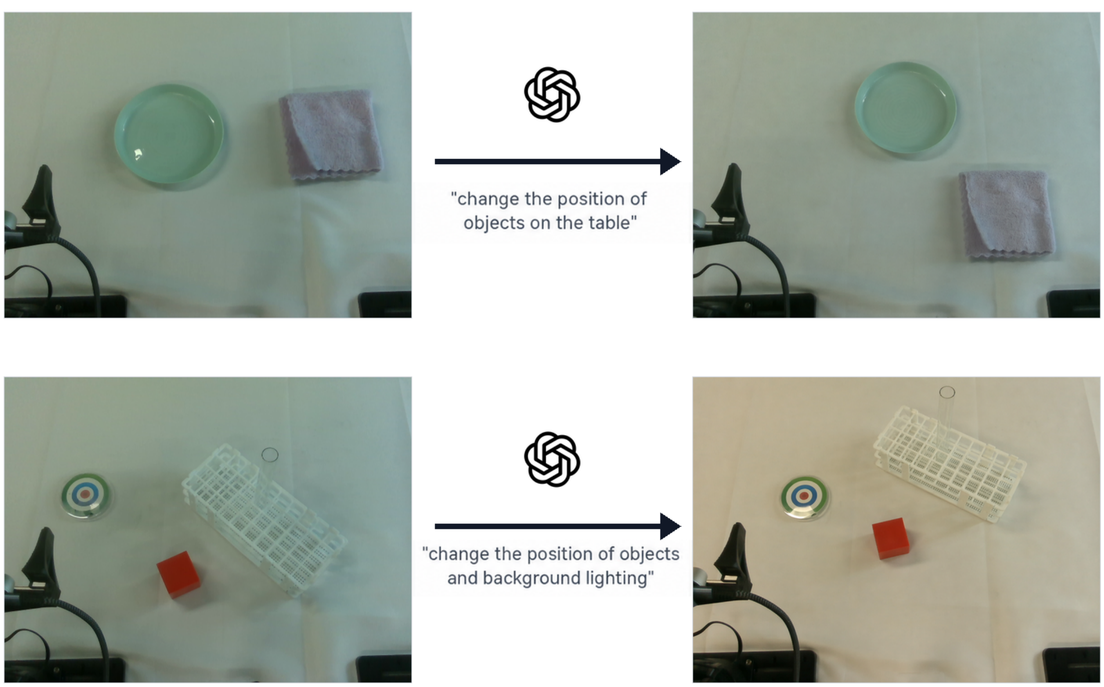

Figure S3: Real-world initial-state augmentation with GPT-Image-2. In each row, the collected overhead observation (left) is edited into a new conditioning image (right). The top example changes object placement, while the bottom example changes both object placement and illumination.

region is a black placeholder. The student VAE has a spatial downsampling factor of 16, so this image is represented on a 28×28 latent grid. Consequently, latent rows $h \in [0, 14)$ and all columns $w \in [0, 28)$ are valid, whereas rows $h \in [14, 28)$ are ignored. Multi-view samples that do not contain an explicit spatial mask are assigned an all-one mask and therefore retain the original full-image loss.

Temporal masking. Video clips are padded to a fixed training length before VAE encoding. Because the student VAE downsamples time by a factor of four, frame-level padding flags are converted into a latent-time validity mask. A latent step is ignored only when all contributing video frames are padding, and the video loss is averaged over the remaining valid latent steps. The first frame, which provides the image condition, is handled separately. This temporal masking is the same for single-view and multi-view samples; single-view pseudo videos additionally use the spatial mask described above. Action chunks are fitted to the student horizon, with padded action positions excluded from supervision.

Rollout-duration cutoff. Simulation pseudo-rollouts are truncated to the first 8 seconds of video before constructing pseudo-data to mitigate long-horizon generation errors. Real-world pseudo-rollouts use all frames from the 20-second generated video and do not apply this temporal cutoff, which improves real-world task success rates. This duration cutoff is separate from the padding-based temporal mask above.

D Full RoboTwin Task-Level Results

Tables S14 and S15 expand the main paper's aggregate RoboTwin results to all 50 tasks under clean and randomized evaluation settings. Vid2WAM achieves the best performance on low-data and novel regimes under both evaluation conditions. Table S15 also marks the 15 withheld target tasks in bold. Vid2WAM leads both full-benchmark and novel-subset aggregates under clean and randomized settings, with larger gains on the novel subset.

Task	Vid2WAM		Fast-WAM		$\pi_{0.5}$		Motus	
	Clean	Random	Clean	Random	Clean	Random	Clean	Random
adjust_bottle	0.95	**0.90**	**0.99**	0.86	0.94	0.88	0.85	0.87
beat_block_hammer	**0.84**	**0.66**	0.69	0.52	0.75	0.64	0.48	0.29
blocks_ranking_rgb	**0.42**	**0.16**	0.39	0.13	0.13	0.15	0.27	**0.16**
blocks_ranking_size	**0.34**	0.11	0.16	0.08	0.10	0.09	0.19	**0.17**
click_alarmclock	0.82	0.69	0.78	0.51	0.79	0.84	**0.93**	**0.94**
click_bell	0.96	0.84	0.95	0.85	0.92	**0.94**	**0.97**	0.88
dump_bin_bigbin	**0.90**	**0.77**	0.88	0.74	0.69	0.47	0.62	0.51
grab_roller	**0.99**	**0.91**	**0.99**	0.89	0.92	0.74	0.98	0.73
handover_block	**0.30**	**0.20**	0.13	0.05	0.05	0.04	0.09	0.02
handover_mic	**0.96**	**0.63**	0.95	0.62	0.32	0.20	0.46	0.44
lift_pot	0.14	0.17	0.18	0.20	**0.59**	**0.39**	0.14	0.09
move_can_pot	0.49	**0.45**	0.57	0.33	**0.60**	0.41	0.08	0.25
move_playingcard_away	0.78	0.68	0.73	0.59	0.81	0.86	**0.92**	**0.87**
move_stapler_pad	0.03	0.02	0.01	0.00	**0.18**	**0.11**	0.08	0.04
hanging_mug	0.07	**0.11**	0.09	0.10	0.02	0.03	**0.17**	0.06
open_laptop	0.82	0.69	**0.84**	0.63	0.71	**0.72**	0.70	**0.72**
open_microwave	0.30	0.18	0.24	0.18	0.05	0.09	**0.57**	**0.40**
pick_diverse_bottles	0.25	0.11	0.20	0.03	0.44	0.23	**0.45**	**0.33**
pick_dual_bottles	0.29	0.15	0.25	0.12	**0.59**	**0.51**	0.58	0.38
place_a2b_left	0.30	0.26	0.24	0.09	0.49	0.42	**0.77**	**0.60**
place_a2b_right	0.34	0.33	0.21	0.14	0.45	0.37	**0.77**	**0.57**
place_bread_basket	**0.70**	0.56	0.60	0.51	0.61	**0.58**	0.54	0.54
place_bread_skillet	0.45	0.29	0.55	0.23	0.36	0.24	**0.57**	**0.44**
place_can_basket	0.68	0.52	**0.79**	0.51	0.62	**0.54**	0.29	0.32
place_cans_plasticbox	**0.88**	**0.69**	**0.88**	**0.69**	0.52	0.42	0.28	0.18
place_container_plate	0.96	**0.92**	**0.98**	0.87	0.95	0.82	0.91	0.79
place_dual_shoes	0.15	0.11	**0.22**	**0.13**	0.12	0.06	0.19	0.02
place_empty_cup	0.82	0.64	0.69	0.48	0.77	0.80	**0.88**	**0.86**
place_fan	0.18	0.13	**0.30**	0.13	0.08	0.14	0.18	**0.15**
place_burger_fries	**0.84**	0.77	0.79	0.69	0.81	**0.82**	0.57	0.58
place_mouse_pad	0.18	0.07	0.09	0.03	**0.28**	0.14	0.18	**0.29**
place_object_basket	0.70	0.59	**0.72**	**0.69**	0.31	0.17	0.59	0.61
place_object_scale	0.36	0.28	0.24	0.13	0.45	**0.42**	**0.49**	0.35
place_object_stand	**0.88**	**0.80**	0.81	0.52	0.75	0.61	0.60	0.56
place_phone_stand	**0.60**	**0.60**	0.51	0.36	0.47	0.47	0.38	0.20
move_pillbottle_pad	0.37	0.22	0.31	0.20	**0.54**	**0.58**	0.36	0.15
place_shoe	0.37	0.21	0.37	0.34	0.24	0.21	**0.57**	**0.61**
press_stapler	0.92	0.81	0.87	0.68	0.80	0.71	**1.00**	**0.91**
put_bottles_dustbin	**0.13**	**0.15**	0.08	0.04	0.03	0.01	0.07	0.00
put_object_cabinet	**0.34**	**0.22**	0.29	0.20	0.09	0.06	0.10	0.19
rotate_qrcode	0.53	**0.33**	**0.54**	0.22	0.32	0.21	0.24	0.22
scan_object	0.31	0.21	0.26	0.24	0.28	0.25	**0.59**	**0.49**
shake_bottle	**1.00**	**1.00**	**1.00**	0.98	0.96	0.97	0.94	0.94
shake_bottle_horizontally	**1.00**	**1.00**	**1.00**	0.97	0.97	0.97	0.93	0.91
stack_blocks_three	**0.30**	**0.18**	0.18	0.10	0.05	0.07	0.00	0.03
stack_blocks_two	0.66	**0.54**	0.68	0.43	0.60	0.42	**0.70**	**0.54**
stack_bowls_three	**0.62**	**0.54**	0.55	0.39	0.27	0.13	0.31	0.22
stack_bowls_two	**0.89**	**0.85**	0.81	0.69	0.73	0.71	0.79	0.80
stamp_seal	0.35	0.25	0.25	0.16	**0.41**	**0.43**	0.10	0.09
turn_switch	0.22	0.18	0.24	0.26	0.40	0.31	**0.54**	**0.49**
Overall	**0.5536**	**0.4536**	0.5214	0.3906	0.4866	0.4280	0.4992	0.4360

Table S14: Success rates on the RoboTwin benchmark (low-data regime).

Task	Vid2WAM		Fast-WAM		$\pi_{0.5}$		Motus	
	Clean	Random	Clean	Random	Clean	Random	Clean	Random
adjust_bottle	**1.00**	**1.00**	**1.00**	0.99	0.99	0.98	**1.00**	**1.00**
beat_block_hammer	0.99	0.96	**1.00**	**0.99**	0.89	0.89	0.83	0.85
blocks_ranking_rgb	**0.08**	0.02	0.06	**0.08**	0.03	0.04	0.04	**0.08**
blocks_ranking_size	0.83	0.84	**0.85**	0.82	0.69	0.42	0.79	**0.87**
click_alarmclock	0.55	**0.81**	0.50	0.44	**0.58**	0.53	0.49	0.79
click_bell	**1.00**	**1.00**	**1.00**	**1.00**	0.91	0.79	**1.00**	**1.00**
dump_bin_bigbin	**0.98**	**0.94**	**0.98**	0.93	0.88	0.89	0.94	**0.94**
grab_roller	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	0.99	0.99
handover_block	**0.95**	0.86	**0.95**	0.85	0.78	0.72	0.92	**0.89**
handover_mic	**1.00**	**1.00**	0.99	0.99	0.97	0.83	**1.00**	**1.00**
lift_pot	0.98	0.99	**0.99**	**1.00**	0.80	0.86	0.97	0.96
move_can_pot	0.94	0.98	**0.98**	**0.99**	0.48	0.56	0.93	0.97
move_playingcard_away	0.04	0.02	**0.11**	0.00	0.08	**0.08**	0.06	0.01
move_stapler_pad	0.72	**0.73**	0.67	0.65	0.63	0.55	**0.77**	0.71
hanging_mug	**0.66**	**0.51**	0.49	0.47	0.08	0.12	0.51	0.44
open_laptop	**0.18**	**0.14**	0.04	0.03	0.12	**0.14**	0.11	0.10
open_microwave	0.47	0.48	0.55	0.55	0.53	0.65	**0.77**	**0.72**
pick_diverse_bottles	0.68	0.76	**0.78**	**0.86**	0.74	0.66	0.48	0.61
pick_dual_bottles	0.72	0.76	**0.99**	**0.91**	0.79	0.81	0.65	0.72
place_a2b_left	**0.96**	0.93	0.94	**0.94**	0.90	0.87	0.94	0.92
place_a2b_right	0.10	**0.17**	0.03	0.10	**0.16**	0.15	0.15	0.15
place_bread_basket	**0.71**	**0.66**	0.01	0.00	0.19	0.21	0.29	0.23
place_bread_skillet	**0.96**	0.85	0.95	**0.95**	0.78	0.79	0.89	0.87
place_can_basket	0.67	0.70	0.71	0.58	**0.74**	**0.73**	0.64	0.72
place_cans_plasticbox	**0.99**	**0.98**	0.98	0.96	0.82	0.75	0.98	0.97
place_container_plate	**0.97**	0.90	0.73	0.61	0.81	0.86	0.95	**0.92**
place_dual_shoes	0.88	**0.93**	**0.92**	0.91	0.68	0.71	0.81	0.85
place_empty_cup	**1.00**	**1.00**	0.99	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
place_fan	**0.97**	**0.96**	**0.97**	0.95	0.83	0.85	**0.97**	0.94
place_burger_fries	**0.17**	0.17	0.02	0.00	0.03	0.02	0.16	**0.21**
place_mouse_pad	0.90	0.86	**0.94**	**0.93**	0.67	0.74	0.89	0.92
place_object_basket	0.74	0.86	0.81	0.81	0.78	0.82	**0.90**	**0.91**
place_object_scale	0.90	0.94	0.89	**0.96**	0.77	0.75	**0.92**	0.93
place_object_stand	**0.62**	**0.63**	0.29	0.34	0.24	0.22	0.41	0.52
place_phone_stand	**0.98**	**0.99**	**0.98**	**0.99**	0.83	0.87	0.89	0.90
move_pillbottle_pad	0.11	0.05	0.01	0.01	**0.18**	**0.15**	0.05	0.04
place_shoe	0.94	**1.00**	0.97	0.99	0.93	0.94	**0.98**	0.99
press_stapler	0.99	**1.00**	**1.00**	0.97	0.94	0.93	**1.00**	**1.00**
put_bottles_dustbin	**0.83**	**0.93**	0.80	0.89	0.49	0.42	0.56	0.72
put_object_cabinet	0.93	**0.91**	**0.94**	**0.91**	0.76	0.77	0.88	0.86
rotate_qrcode	0.93	0.85	**0.96**	**0.90**	0.81	0.84	0.89	0.85
scan_object	**0.94**	0.89	0.91	**0.95**	0.73	0.64	0.82	0.91
shake_bottle	**1.00**	**1.00**	**1.00**	0.99	0.99	0.97	**1.00**	**1.00**
shake_bottle_horizontally	**1.00**	0.99	**1.00**	0.99	**1.00**	**1.00**	**1.00**	**1.00**
stack_blocks_three	**0.97**	0.95	0.96	**0.96**	0.86	0.84	0.94	0.91
stack_blocks_two	**1.00**	**1.00**	**1.00**	0.97	0.97	0.98	**1.00**	**1.00**
stack_bowls_three	0.81	**0.82**	0.77	0.80	**0.84**	0.66	0.83	0.74
stack_bowls_two	0.96	**0.97**	0.96	**0.97**	0.86	0.82	**0.98**	0.96
stamp_seal	0.77	**0.86**	**0.84**	0.77	0.71	0.73	0.71	0.65
turn_switch	0.68	0.69	0.65	0.72	0.69	0.69	**0.86**	**0.90**
Overall	**0.7830**	**0.7848**	0.7572	0.7474	0.6792	0.6648	0.7508	0.7628
Novel subset	**0.5467**	**0.5527**	0.4500	0.4280	0.4653	0.4627	0.4893	0.5153

Table S15: Success rates on the RoboTwin benchmark (novel regime). Novel target tasks are marked in bold.